# AI Drawing Control System

Project Explanation | Timeline: 12 weeks (05 Oct - 24 Dec 2026) | Team: 3 developers

## 1. Project Objective

A web system to store, control and approve engineering drawings product-wise. Every new or revised drawing must be viewed and approved by all listed users before it is released. Every decision is recorded, and old revisions are locked so only the latest approved drawing is in use.

## 2. Key Features

| | |
|---|---|
| **Product Tree** | Drawings are organised in a tree structure, product-wise. |
| **Drawing Upload** | Admin uploads drawings as controlled PDF copies only. Any other file format is rejected by the system. |
| **Version Control** | If a drawing with the same name is uploaded, the previous version is locked. It is kept in the system, but only the admin can access it. |
| **Email Notification** | Every user on the Email page receives a mail for each new or updated drawing. Reminder mails continue until the user responds. |
| **On-Screen Review** | The user views the drawing on screen (download optional) and clicks Confirm or Deny on the same screen. |
| **Deny with Reason** | Deny requires a reason. It is sent to the admin and visible to all users. |
| **All-User Approval** | A drawing becomes Approved only when every user confirms. If anyone denies, the admin uploads a revision and the cycle restarts. |
| **Audit Record** | Saved permanently for every drawing and revision: who confirmed or denied, when, and why. |
| **Download Log** | Every download is recorded with the user name, drawing name, revision, date and time. The admin can view and filter this log by user, drawing or date. |

## 3. Approval Workflow

Upload drawing → mail sent to all users → each user views on screen → Confirm or Deny (with reason) → any Deny goes to the admin and is shown to all users → admin uploads revision (old version locked) → cycle repeats → **Approved** when all users confirm.

## 4. Timeline (12 Weeks)

| Phase | Weeks | Target Date | Main Work |
|---|---|---|---|
| **1. Planning & Design** | W1 | 09 Oct 2026 | Workflow, roles, wireframes, tech stack |
| **2. Core Setup** | W2-3 | 23 Oct 2026 | Login, roles, database, file storage, product tree |
| **3. Upload & Versioning** | W3-4 | 30 Oct 2026 | Controlled PDF upload, same-name detection, locking old versions |
| **4. On-Screen Viewer** | W4-6 | 13 Nov 2026 | PDF viewer, download, Confirm/Deny on screen |
| **5. Approval & Email** | W5-8 | 27 Nov 2026 | Auto mails, reminders, deny reasons, all-user approval rule |
| **6. Admin Panel & Audit** | W8-9 | 04 Dec 2026 | User/email management, locked versions, approval records, download log |
| **7. Testing & UAT** | W10-11 | 18 Dec 2026 | Functional, PDF viewing, email and security testing; user testing |
| **8. Deploy & Training** | W12 | 24 Dec 2026 | Server setup, upload existing drawings, training, go-live |

## 5. Key Decisions & Risks

- **Controlled PDF only:** The system accepts, shows and downloads only controlled PDF copies, so drawings open on screen in any browser without extra viewer software.
- **Team size:** 3 months needs 3 developers. With 2 developers, expect 14-15 weeks.
- **AI features (after go-live):** automatic comparison of old vs new revision, title-block reading, and smart search.